

## LA CORUNA AP, Spain

WMO: 080020

Lat: 43.307N

Lon: 8.372W

Elev: 99

StdP: 100.15

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 0.1        | 1.5 | -2.1                          | 3.2 | 4.1  | -0.8 | 3.6 | 4.6  | 12.7                  | 12.3 | 11.1 | 11.7 | 2.2                   | 180  | 0.454 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.5                    | 28.8            | 20.5 | 26.8 | 19.5 | 25.0 | 18.9 | 21.5                | 26.5 | 20.6 | 24.8 | 19.8 | 23.5 | 3.4                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.0                            | 14.9 | 23.5 | 19.1 | 14.0 | 22.6 | 18.2 | 13.3 | 21.9 | 63.1          | 26.9 | 60.0 | 24.9 | 57.4 | 23.5 | 25.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.4           | 8.2  | 7.2 | -1.9                       | 32.6 | 1.1                | 1.8 | -2.7                                               | 33.9 | -3.3       | 34.9 | -3.9       | 35.9 | -4.7       | 37.2 |    |    |
| (5)               |      |     | -2.2                       | 22.8 | 1.2                | 1.2 | -3.0                                               | 23.7 | -3.7       | 24.4 | -4.4       | 25.1 | -5.2       | 26.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 14.3        | 9.9   | 9.8   | 11.0  | 12.5  | 14.9  | 17.5  | 19.5  | 19.5  | 18.5  | 15.9  | 12.2  | 10.6  |  |  |
|                                                                        | DBStd       | 4.43        | 2.89  | 2.58  | 2.50  | 2.67  | 2.43  | 2.30  | 2.28  | 2.07  | 2.47  | 2.71  | 2.87  | 2.90  |  |  |
|                                                                        | HDD10.0     | 134         | 38    | 32    | 18    | 6     | 0     | 0     | 0     | 0     | 0     | 1     | 11    | 29    |  |  |
|                                                                        | HDD18.3     | 1616        | 261   | 238   | 229   | 177   | 111   | 43    | 12    | 12    | 27    | 82    | 184   | 240   |  |  |
|                                                                        | CDD10.0     | 1718        | 36    | 27    | 47    | 81    | 153   | 224   | 293   | 294   | 255   | 184   | 77    | 47    |  |  |
|                                                                        | CDD18.3     | 158         | 0     | 0     | 0     | 1     | 4     | 17    | 47    | 48    | 32    | 8     | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 814         | 0     | 0     | 3     | 12    | 49    | 109   | 236   | 212   | 157   | 35    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 177         | 0     | 0     | 0     | 3     | 9     | 23    | 69    | 36    | 31    | 6     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.1         | 3.2   | 3.5   | 3.5   | 3.3   | 3.3   | 3.1   | 3.0   | 2.9   | 2.6   | 2.8   | 3.1   | 3.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1142        | 136   | 93    | 90    | 106   | 78    | 46    | 37    | 42    | 69    | 150   | 168   | 127   |  |  |
|                                                                        | PrecMax     | 1479        | 265   | 180   | 308   | 316   | 190   | 119   | 114   | 97    | 161   | 337   | 316   | 303   |  |  |
|                                                                        | PrecMin     | 788         | 33    | 2     | 0     | 26    | 19    | 0     | 1     | 0     | 10    | 14    | 21    | 21    |  |  |
|                                                                        | PrecStd     | 190         | 67    | 55    | 75    | 64    | 39    | 32    | 28    | 27    | 44    | 76    | 78    | 62    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 17.2  | 20.0  | 23.1  | 24.2  | 27.9  | 29.7  | 32.1  | 29.9  | 30.1  | 26.2  | 22.0  | 18.3  |  |  |
|                                                                        |             | MCWB        | 14.3  | 12.4  | 13.3  | 16.2  | 18.8  | 20.0  | 22.9  | 21.7  | 21.0  | 17.9  | 16.4  | 13.8  |  |  |
|                                                                        | 2%          | DB          | 16.1  | 17.0  | 19.2  | 21.6  | 24.2  | 26.2  | 28.2  | 27.2  | 26.9  | 23.9  | 19.2  | 17.1  |  |  |
|                                                                        |             | MCWB        | 13.9  | 11.6  | 12.4  | 15.2  | 16.9  | 18.6  | 21.2  | 20.2  | 19.4  | 17.4  | 15.8  | 13.7  |  |  |
|                                                                        | 5%          | DB          | 15.2  | 15.2  | 17.1  | 19.6  | 21.3  | 24.0  | 25.7  | 25.8  | 25.0  | 22.1  | 17.9  | 16.0  |  |  |
|                                                                        |             | MCWB        | 13.1  | 11.8  | 11.8  | 14.2  | 16.0  | 18.3  | 20.0  | 19.8  | 18.8  | 16.9  | 14.8  | 13.2  |  |  |
|                                                                        | 10%         | DB          | 14.2  | 14.1  | 15.2  | 17.7  | 19.2  | 21.9  | 23.9  | 24.0  | 23.2  | 20.7  | 16.2  | 15.1  |  |  |
|                                                                        |             | MCWB        | 12.1  | 11.3  | 11.6  | 13.4  | 14.8  | 17.3  | 19.2  | 19.2  | 18.2  | 16.7  | 14.0  | 12.6  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 15.5  | 14.0  | 14.8  | 17.0  | 19.1  | 21.2  | 23.4  | 22.6  | 22.0  | 19.7  | 18.0  | 15.6  |  |  |
|                                                                        |             | MCDB        | 16.6  | 16.6  | 19.1  | 23.3  | 25.7  | 27.9  | 31.1  | 27.4  | 28.1  | 23.2  | 19.9  | 17.0  |  |  |
|                                                                        | 2%          | WB          | 14.4  | 13.1  | 13.6  | 15.7  | 17.6  | 19.7  | 21.6  | 21.2  | 20.7  | 18.7  | 16.6  | 14.6  |  |  |
|                                                                        |             | MCDB        | 15.8  | 15.6  | 17.2  | 20.0  | 23.2  | 25.0  | 27.0  | 25.3  | 24.6  | 21.9  | 18.8  | 16.1  |  |  |
|                                                                        | 5%          | WB          | 13.3  | 12.4  | 12.9  | 14.8  | 16.5  | 18.5  | 20.5  | 20.5  | 19.9  | 17.9  | 15.5  | 13.8  |  |  |
|                                                                        |             | MCDB        | 14.9  | 15.0  | 16.0  | 18.7  | 20.5  | 22.8  | 24.6  | 24.1  | 23.3  | 20.9  | 17.3  | 15.6  |  |  |
|                                                                        | 10%         | WB          | 12.6  | 11.5  | 12.1  | 13.9  | 15.4  | 17.6  | 19.5  | 19.7  | 19.0  | 17.1  | 14.4  | 13.0  |  |  |
|                                                                        |             | MCDB        | 14.2  | 14.0  | 15.2  | 17.2  | 18.7  | 21.0  | 23.0  | 23.0  | 22.4  | 20.0  | 16.2  | 14.8  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.5   | 7.6   | 8.2   | 8.5   | 7.9   | 7.6   | 7.7   | 8.5   | 9.2   | 8.9   | 7.0   | 7.3   |  |  |
|                                                                        |             | MCDBR       | 6.2   | 10.2  | 12.2  | 13.0  | 12.5  | 12.3  | 11.8  | 11.9  | 12.5  | 11.6  | 8.7   | 7.6   |  |  |
|                                                                        | 5% WB       | MCWBR       | 4.5   | 6.5   | 7.0   | 7.5   | 6.9   | 6.3   | 5.7   | 5.9   | 6.0   | 6.3   | 5.7   | 5.2   |  |  |
|                                                                        |             | MCWBR       | 5.2   | 8.1   | 9.8   | 11.3  | 10.9  | 10.2  | 10.1  | 9.9   | 9.2   | 7.1   | 5.6   | 5.6   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.336 | 0.342 | 0.373 | 0.380 | 0.383 | 0.387 | 0.376 | 0.375 | 0.368 | 0.363 | 0.343 | 0.330 |  |  |
|                                                                        |             | taud        | 2.475 | 2.452 | 2.372 | 2.360 | 2.381 | 2.390 | 2.426 | 2.442 | 2.459 | 2.478 | 2.474 | 2.487 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 793   | 853   | 865   | 886   | 890   | 884   | 890   | 881   | 861   | 818   | 775   | 764   |  |  |
|                                                                        |             | Edh at Noon | 74    | 88    | 107   | 117   | 118   | 117   | 112   | 107   | 97    | 85    | 73    | 67    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.43  | 2.34  | 3.50  | 4.74  | 5.68  | 6.20  | 6.21  | 5.57  | 4.44  | 2.79  | 1.69  | 1.32  |  |  |
|                                                                        | RadStd      | RadStd      | 0.17  | 0.34  | 0.51  | 0.47  | 0.49  | 0.42  | 0.37  | 0.41  | 0.27  | 0.32  | 0.24  | 0.17  |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page